



Lovely Professional University

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

PROJECT REPORT

(INT387 – Data Analysis and Visualization Project)

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Topic:

EXIM Bank Trade Finance Dashboard

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DECLARATION

We, Khushi and Ansh Batra, students of Bachelor of Technology under the CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on our own intensive work and is genuine.

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COURSE COMPLETION CERTIFICATE

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1. Introduction

International trade is a backbone of economic growth, and financial institutions such as EXIM Bank play a key role in enabling exports and imports through structured financing. Trade finance mitigates risks such as delayed payments, currency fluctuations, and political instability, allowing exporters and importers to transact with confidence across borders.

This project analyses EXIM Bank's trade finance data using Power BI to uncover patterns in authorization, disbursement, and regional distribution. The resulting dashboard converts complex financial records into an interactive, visual format that allows stakeholders to quickly understand performance trends and drill into specific programs, countries, or states.

The overarching aim is to equip decision-makers with actionable insights that improve operational efficiency, optimize resource allocation, and strengthen the bank's global trade strategy.

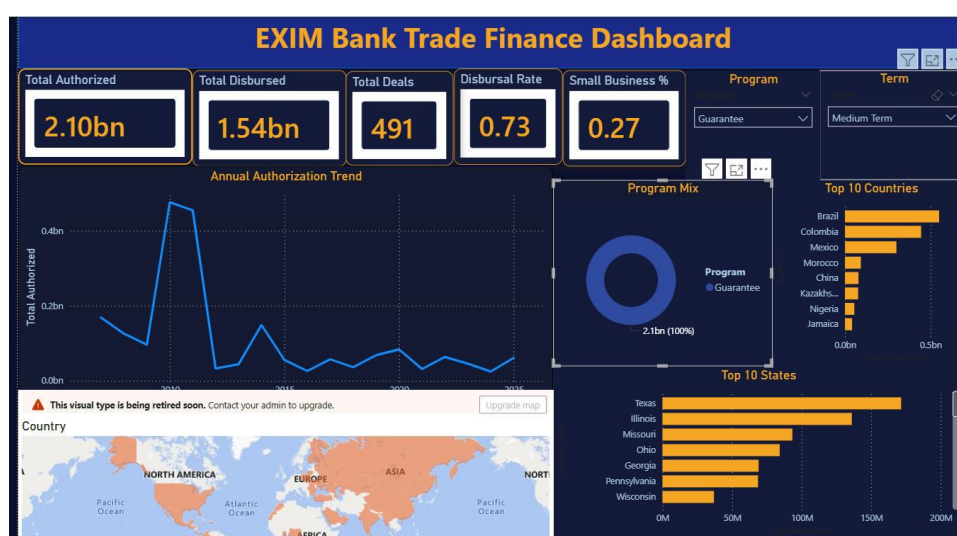


Fig. 1.1 – EXIM Bank Trade Finance Dashboard Overview

2. Objectives

The primary objective of this project is to convert raw financial data into meaningful insights using visualization techniques. Specifically, the project aims to:

- Analyse authorization trends over time and identify periods of growth and decline.
- Evaluate disbursement efficiency and overall fund utilization.
- Identify top-performing countries and states by authorized trade finance value.
- Assess the contribution of small businesses to the overall trade finance portfolio.
- Understand how funding is distributed across different financing programs (Guarantee, Insurance, Loan, and Working Capital).
- Enable dynamic, filter-driven exploration of the dataset through an interactive dashboard.

In doing so, the project also strengthens data-interpretation and visual-storytelling skills and simulates a real-world business-intelligence scenario in which data directly informs strategic decision-making.

The dataset EXIM_Bank_data.xlsx contains transaction-level trade finance records and includes the following key fields:

- Total Authorized Amount
- Total Disbursed Amount
- Number of Deals
- Program Type (Guarantee, Insurance, Loan, Working Capital)
- Country and State
- Business Segment (including Small Business indicator)
- Time-based Data (Fiscal Year, Decision Date, Expiration Date)

The dataset combines quantitative fields, such as authorized and disbursed amounts, with categorical fields, such as program type, country, and state, making it well suited to multidimensional analysis. Each record represents an individual transaction or deal, which makes the data readily aggregable for KPI calculation and trend analysis.

The data spans multiple fiscal years, enabling year-over-year trend analysis, and reflects real-world financial patterns that make the resulting insights practical and industry-relevant. Overall, the dataset is well suited to business intelligence work, as it combines financial, geographical, and categorical dimensions in a single structure.

[illegible]

Fig. 3.1 – EXIM Bank data.xlsx in Microsoft Excel

4. Data Preprocessing

Before the data was loaded into Power BI, the following preprocessing steps were carried out to ensure accuracy and consistency:

- Removed records with missing or incomplete values.
- Cleaned inconsistent entries in categorical fields such as country and program type.
- Standardized column names for clarity and consistency across the model.
- Converted date fields to a proper date format and extracted a Fiscal Year column for trend analysis.
- Removed duplicate records to preserve data integrity.
- Structured and validated the data for direct import into Power BI.

Outliers were also reviewed to ensure they did not distort the resulting KPIs and visualizations. Together, these steps ensured that the dataset used for analysis was clean, consistent, and analysis-ready.

5. Tools & Technologies

- Power BI Desktop – primary tool for interactive data visualization.
- Microsoft Excel – used for initial data handling, inspection, and cleaning.
- Power Query – used within Power BI for data transformation and filtering.
- DAX (Data Analysis Expressions) – used to build calculated measures and KPIs.



Fig. 5.1 – Power BI Desktop and Microsoft Excel

Power BI was used as the primary visualization platform because of its powerful interactive capabilities, while Microsoft Excel supported initial data handling and inspection. Within Power BI, data-transformation techniques such as filtering, aggregation, and DAX-based calculated measures were applied, and visualizations including bar charts, line charts, donut charts, and maps were used to represent the underlying insights effectively.

6. Dashboard Overview

The dashboard was designed to provide a clear, interactive view of EXIM Bank's trade finance data, and includes:

- KPI cards summarizing key performance metrics.
- A line chart tracking the annual authorization trend.
- A donut chart showing the mix of financing programs.
- Bar charts ranking the top countries and states by authorization.
- A geographical map visualizing the global spread of trade finance activity.

At the top of the dashboard, KPI cards display key metrics — total authorized amount, total disbursed amount, and total number of deals — giving users an immediate snapshot of overall performance.



Fig. 6.1 – KPI Cards

Below the KPI cards, multiple visualizations display trends, distributions, and comparisons. Program and Term filters allow users to explore specific segments of the portfolio, making the dashboard dynamic, drill-down capable, and user-friendly. The overall layout ensures that users can quickly grasp headline performance while still being able to explore the underlying detail.

7. KPI Analysis



Fig. 7.1 – Key Performance Indicators

KPI	Value & Interpretation
Total Authorized	₹2.10 Billion — total funds sanctioned by the bank.
Total Disbursed	₹1.54 Billion — funds actually released to borrowers.
Total Deals	491 — total number of trade finance transactions.
Disbursal Rate	0.73 — share of authorized funds that has been disbursed, an efficiency indicator.
Small Business %	0.27 — proportion of authorization attributable to small businesses.

The KPI section provides a quick summary of overall performance. The total authorized amount of ₹2.10 billion represents the total funds sanctioned by the bank, while the disbursed amount of ₹1.54 billion indicates actual fund utilization.

The disbursal rate of 0.73 reflects strong operational efficiency, showing that nearly three-quarters of authorized funds are successfully distributed to borrowers. The total deal count of 491 highlights healthy transaction volume across the portfolio.

The small business percentage of 0.27 is particularly noteworthy, as it reflects the inclusivity of the program and its role in supporting emerging enterprises that might otherwise struggle to access trade finance.

8. Annual Authorization Trend

- Authorization activity rose steadily from 2007, peaking around 2013 at approximately ₹12 billion.
- A sharp decline followed between 2015 and 2018, after which the trend stabilized at a lower level.
- From 2019 to 2024, authorization amounts fluctuated between roughly ₹2 billion and ₹5 billion annually.



Fig. 8.1 – Annual Authorization Trend

The annual trend analysis reveals a pronounced growth phase in authorization amounts through the early 2010s, culminating in a peak around 2013. This was followed by a sustained decline and a subsequent period of lower, fluctuating activity through to 2024.

These fluctuations are likely influenced by global economic conditions, shifts in trade policy, and changing market demand. Identifying such periods of growth and decline helps the bank improve forecasting and long-term planning.

9. Program Mix Analysis

- Working Capital financing accounts for the largest share of authorized funds, at approximately 54.3% (₹55.57 billion).
- Guarantee programs represent the second-largest share, at roughly 27.5% (₹28.13 billion).
- Insurance accounts for around 10.0% (₹10.21 billion), with Loan programs making up the remainder.



Fig. 9.1 – Program Mix

The program mix chart shows how authorized funds are distributed across the bank's different financing instruments. Working Capital support is the dominant program, indicating that the bank's core role is to help exporters fund day-to-day operational needs rather than simply underwriting risk through guarantees.

The substantial share held by Guarantee and Insurance programs also reflects the importance of structured risk management in trade finance decisions. Understanding this distribution helps the bank optimize its product offerings and allocate resources to the programs with the greatest demand.

10. Country-wise Analysis

- Top markets by authorization: Mexico, India, and Ireland.
- Brazil and China also feature prominently among the top ten countries.
- Insight: trade finance activity spans both emerging and developed markets.

Source: EXIM Bank Trade Finance Dashboard

1 states dominate trade finance

Top 10 States

Fig. 10.1 – Top 10 Countries by Authorization

Country-level analysis highlights the geographical distribution of EXIM Bank's trade finance activity. Mexico, India, and Ireland lead the portfolio, with Turkey, Brazil, China, and Luxembourg also appearing among the top ten — reflecting a broad, globally diversified footprint that spans both emerging and developed economies.

This insight helps the bank identify its key markets and highlight potential areas for further expansion, while also demonstrating the breadth of its global outreach and strategic focus.

11. State-wise Analysis

- Top states by authorization: Washington, Texas, and California.
- Pennsylvania and Florida also rank among the top five states.
- Insight: industrial and export-oriented states dominate trade finance activity.



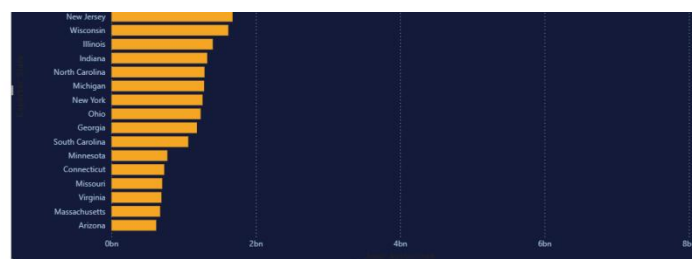
Fig. 11.1 – Top 10 States by Authorization

State-level analysis provides a more granular view of trade finance distribution within the United States. Washington, Texas, and California show the highest authorization amounts, indicating strong export and industrial activity in these states, with Pennsylvania and Florida also contributing significantly.

This breakdown helps identify regional strengths and opportunities and can support more targeted, state-level outreach and policy-making.

12. Visualization Techniques

- Line charts – used to display the annual authorization trend over time.
- Bar charts – used to compare countries and states by authorization value.
- Donut chart – used to show the proportional mix of financing programs.
- Geo map – used to visualize the global distribution of trade finance activity.



(Top 10 States by Authorization)

Fig. 12.1 – Geographical Distribution of Trade Finance Activity

A combination of visualization techniques was used to represent the data as effectively as possible for each type of insight. Line charts were used to communicate trends over time, bar charts for direct comparisons across countries and states, and a donut chart for proportional breakdowns of the program mix.

The geo map, in particular, makes it easy to identify regional patterns at a glance, complementing the more precise, ranked view offered by the bar charts. Together, these visualizations substantially improve data understanding and support faster, better-informed decision-making.

13. Business Impact

- Helps bank stakeholders make faster, better-informed decisions.
- Identifies high-performing countries, states, and financing programs.
- Supports risk assessment and strategic resource allocation.

The dashboard helps stakeholders make informed decisions by providing clear, at-a-glance insight into financial performance across programs, countries, and states. It supports both strategic planning and ongoing risk management.

By identifying key regions and trends, the bank is better positioned to allocate resources effectively, prioritize high-potential markets, and improve overall operational efficiency.

14. Challenges Faced

- Data cleaning complexity, given inconsistent entries across multiple fields.
- Handling missing values without distorting the resulting KPIs.
- Balancing detail and simplicity in the dashboard layout and design.

One of the major challenges was ensuring data accuracy and consistency across a large, multi-year dataset. Designing a dashboard that balances sufficient detail with overall simplicity was also demanding, as was selecting the most appropriate visualization for each type of insight.

15. Limitations

- The analysis is based on a static dataset rather than a live data feed.
- The dashboard does not currently support real-time updates.
- Some advanced analytical features (e.g., predictive modelling) are outside the current scope.

The analysis is limited to the available dataset and may not fully reflect real-time market conditions or external factors that influence trade finance activity. Despite these limitations, the project provides valuable, evidence-based insight drawn from the available data.

16. Conclusion

The EXIM Bank Trade Finance Dashboard demonstrates the value of data visualization in understanding financial performance. By transforming raw transactional data into an interactive Power BI dashboard, the project provides a comprehensive, easy-to-navigate view of trade finance activity across programs, countries, and states, and supports more informed, strategic decision-making by the bank's stakeholders.

17. Future Scope

- Integration of real-time data sources to keep the dashboard continuously up to date.
- Application of AI-based predictive models to forecast future authorization and disbursement trends.
- Incorporation of more advanced analytics, such as risk scoring by country or sector.
- Deployment of the dashboard as a shareable, web-based Power BI report.

Future enhancements may include integrating real-time data sources, applying predictive analytics, and incorporating machine-learning models to anticipate trends rather than simply report on historical performance. The dashboard could also be expanded to include additional variables and more advanced visualizations, and ultimately deployed for wider, web-based access.

18. References

- Power BI Documentation.
- Microsoft Excel Documentation.
- Data Analytics Resources.
- Data.gov – EXIM Bank trade finance dataset.
- ChatGPT (used for guidance and report structuring).